Chebyshev Quadrature Versus Trapezoid Integration

Scalar, random-polynomial, and EuRoC aggressive-snippet evidence

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1 Executive Summary

The evidence supports a conditional, actionable case for Chebyshev2 quadrature. The corrected Chebyshev2 integration path now evaluates the antiderivative in an `N+1` representation, and the regularized runs use the derivative-matrix Tikhonov term with `lambda1 = 0.005`. Those fixes make the comparison fairer, but they still leave one important operating distinction: smooth low-noise signals and noisy aggressive IMU snippets behave differently.

For the practical fixed-size rule `N_sqrt = ceil(sqrt(m))`, the decision view from Sections 3, 4, and 6 is:

- Low-noise synthetic signals: use Chebyshev2 without Tikhonov. analytic scalar: 78% with `lambda1=0`, random degree-4: 80% with `lambda1=0`
- High-noise signals: use Chebyshev2 with Tikhonov. analytic scalar: 100% with `lambda1=0.005`, random degree-4: 84% with `lambda1=0.005`, EuRoC aggressive: 97% with `lambda1=0.005`
- Low-noise EuRoC aggressive snippets are the exception: even the better Chebyshev2 choice wins only 16% with `lambda1=0`, so trapezoid remains the safer low-noise baseline for those underfit, high-curvature snippets.

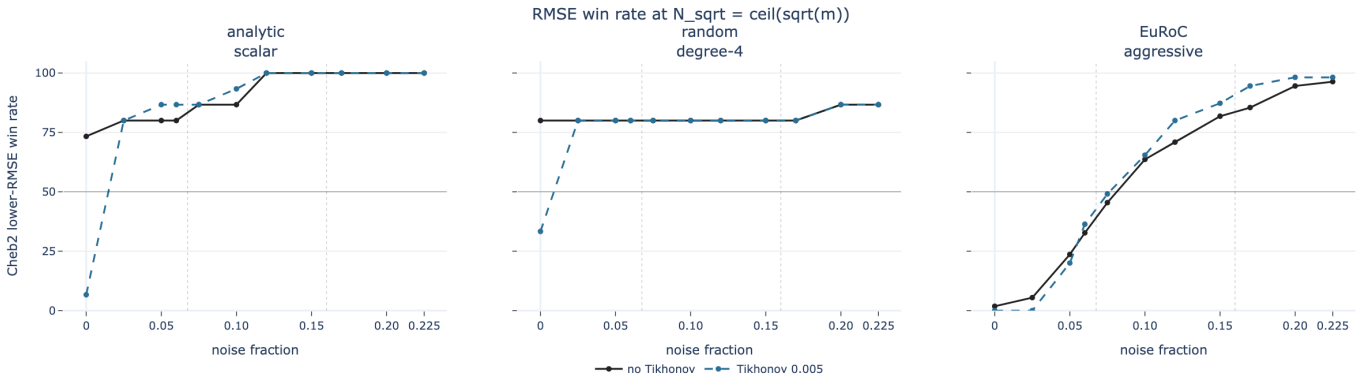


Figure 1: RMSE win rate by noise fraction at `N_sqrt = ceil(sqrt(m))`.

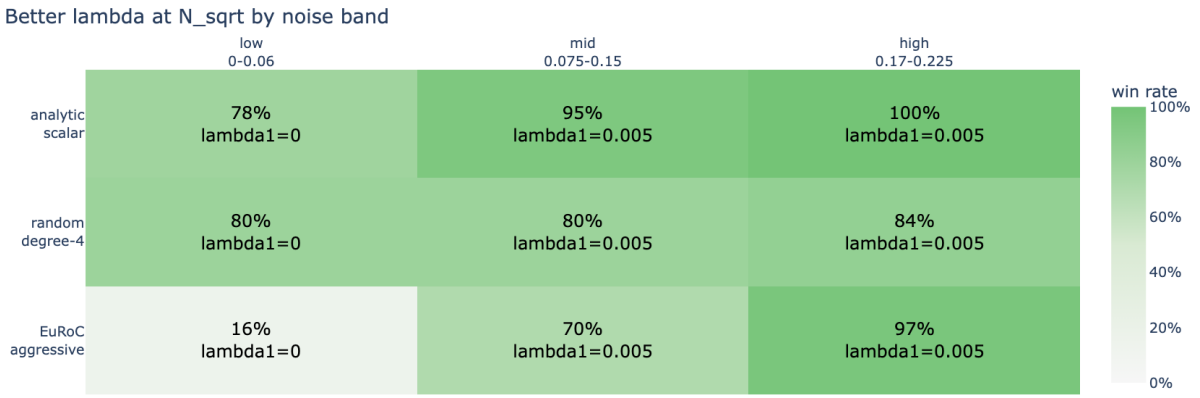


Figure 2: Best Chebyshev2 lambda choice by noise band at `N_sqrt = ceil(sqrt(m))`; cells report RMSE win rate over trapezoid.

The broader pattern is:

- Chebyshev2 is strongest when the signal is smooth and the chosen `N` is modest.
- Trapezoid remains a strong baseline for uniformly spaced low-noise aggressive snippets because it is local and less exposed to global fit artifacts.
- Tikhonov regularization helps most when the Chebyshev2 fit would otherwise use too much high-order freedom. It improves robustness in the high-noise regime, but it should not be applied by default in low-noise synthetic cases.
- The EuRoC snippets are the most relevant stress test. They show a strong high-noise Chebyshev2 win at `N_sqrt`, but they do not support the stronger claim that Chebyshev2 dominates trapezoid in every low-noise aggressive window.

Summary over the decision-view comparisons:

Family	lambda1	RMSE win	Median adv.	Robust-N win	Positive robust rows
analytic scalar functions	0	63%	0.0001375	85%	100%
analytic scalar functions	0.005	65%	0.0001984	88%	100%
random degree-4 polynomials	0	58%	5.902e-05	100%	100%
random degree-4 polynomials	0.005	61%	7.695e-05	96%	100%
EuRoC aggressive 200 ms snippets	0	38%	-2.246e-05	68%	87%
EuRoC aggressive 200 ms snippets	0.005	46%	-9.268e-06	71%	95%

How to read the figures: positive advantage means `trapezoid error - Chebyshev2 error > 0`, so Chebyshev2 has lower error. The black diamond marks the best median RMSE `N`, and the dashed vertical line marks `sqrt(m)`.

2 Reproducing the May 24 Scalar Comparison

The May 24 scalar notebook can be reproduced with the current corrected integration code. The two figures below are deliberately representative old-style image plots: they establish continuity with Benjamin's original scalar comparison, but they are not the best way to choose `N` because the sign, noise dependence, and fixed-`N` slices are hard to compare directly.

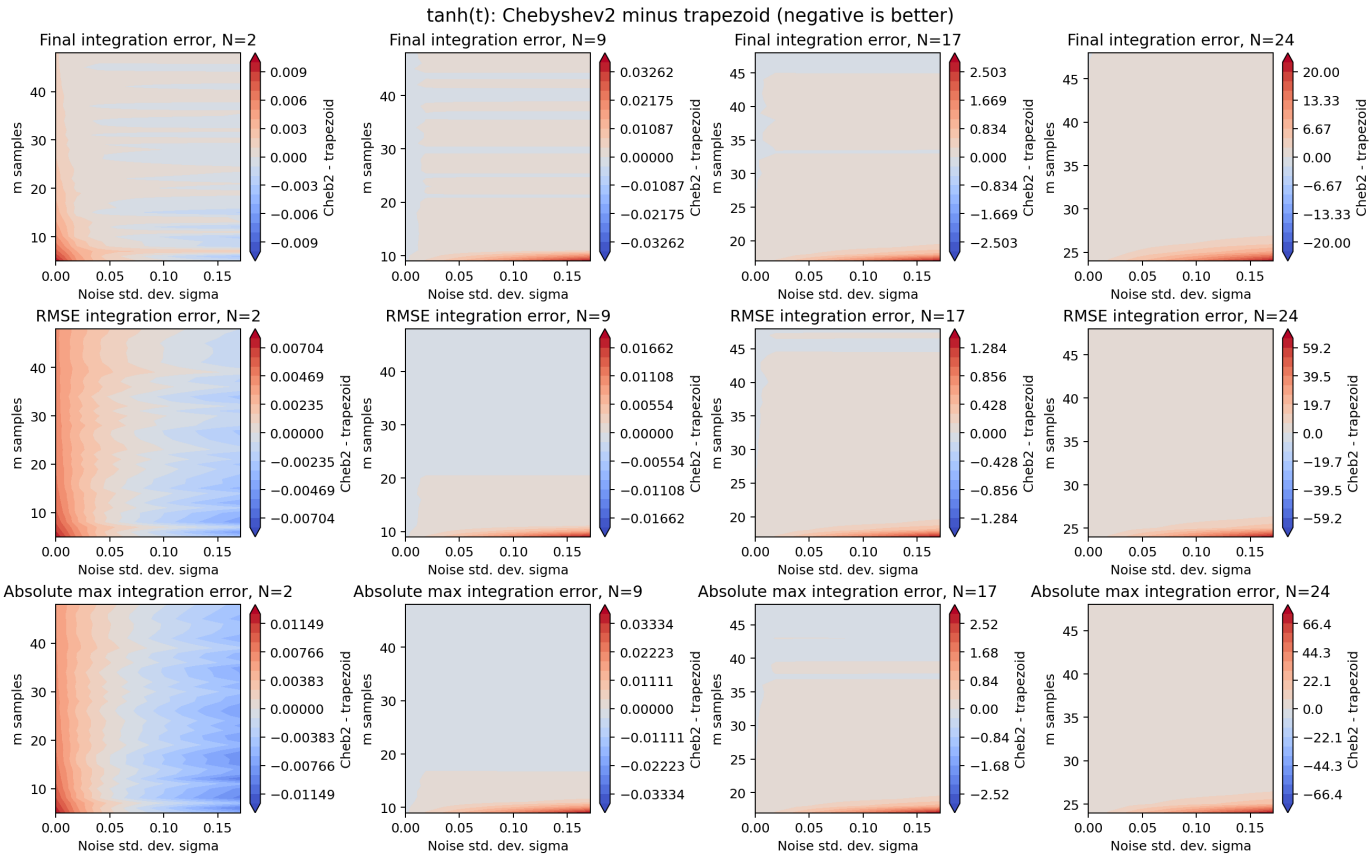


Figure 3: Representative old fixed-N scalar heatmaps for `tanh(t)`.

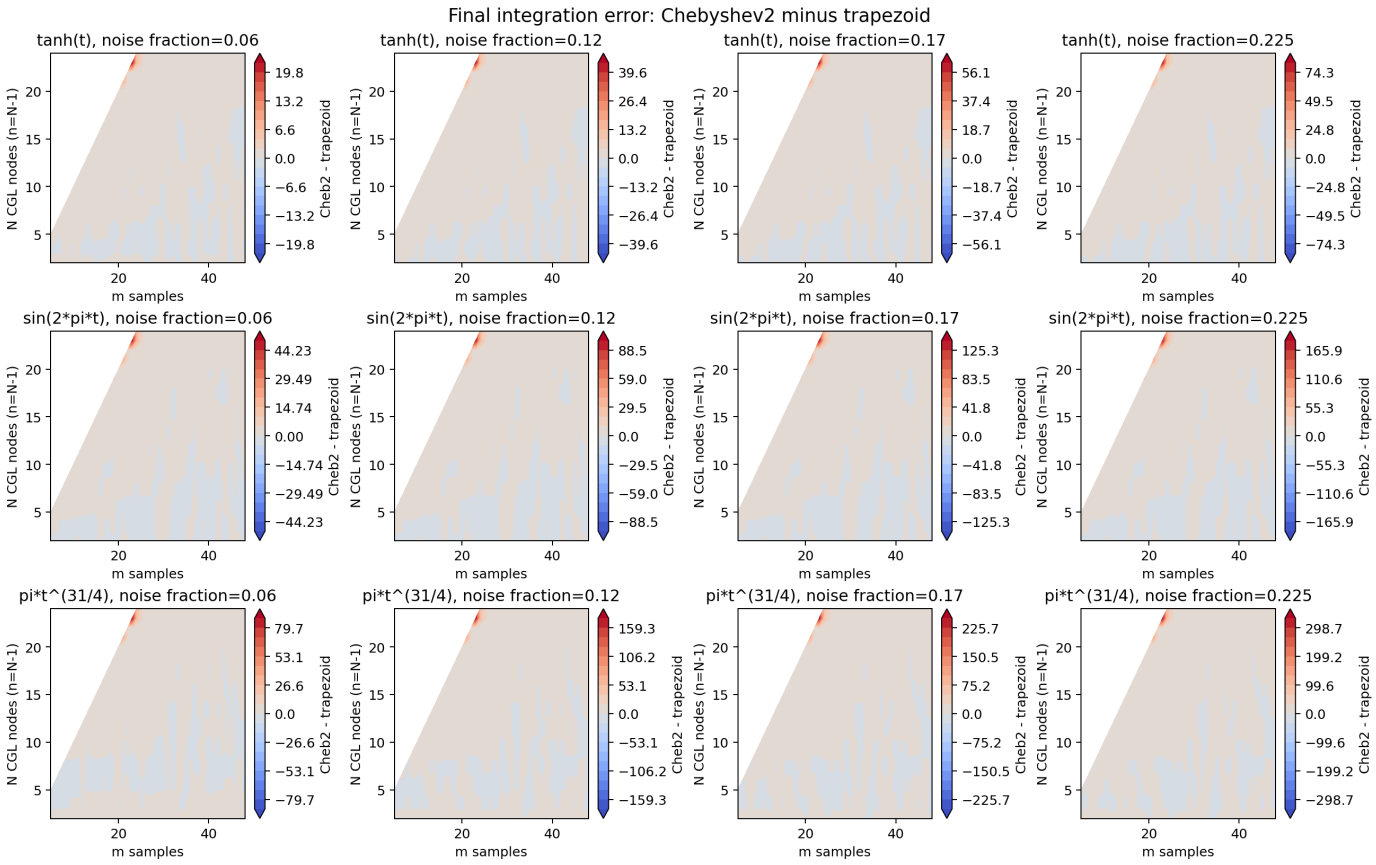


Figure 4: Representative old scalar heatmaps across all three functions at selected noise levels.

3 Decision-oriented analytic scalar views

The same three analytic functions are now shown through the decision-oriented Plotly exports. The unregularized block is the baseline; the regularized block uses lambda1 = 0.005. Positive RMSE advantage means trapezoid error minus Chebyshev2 error is positive, so Chebyshev2 is better.

3.1 No Tikhonov (`lambda1 = 0`)

3.1.1 `tanh(t)`

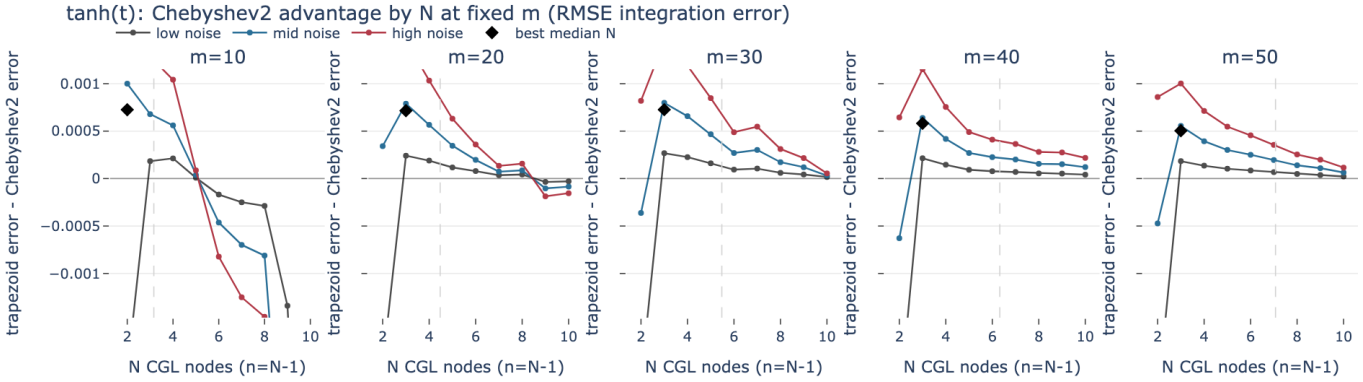


Figure 5: RMSE advantage curves for `tanh(t)` with No Tikhonov (`lambda1 = 0`).

tanh(t): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	2	3	2	0%	75%	100%	0.0007259	@@@@%%#.
20	4.47	6	3	3	3	75%	100%	100%	0.0007155	:@#+-::..
30	5.48	10	3	3	3	75%	100%	100%	0.0007262	.@%#++++=
40	6.32	9	3	3	3	75%	100%	100%	0.0005816	.@%###****
50	7.07	7	3	3	3	75%	100%	100%	0.000504	.@%###****

Figure 6: Robust `N` table for `tanh(t)` with No Tikhonov (`lambda1 = 0`).

3.1.2 `sin(2*pi*t)`

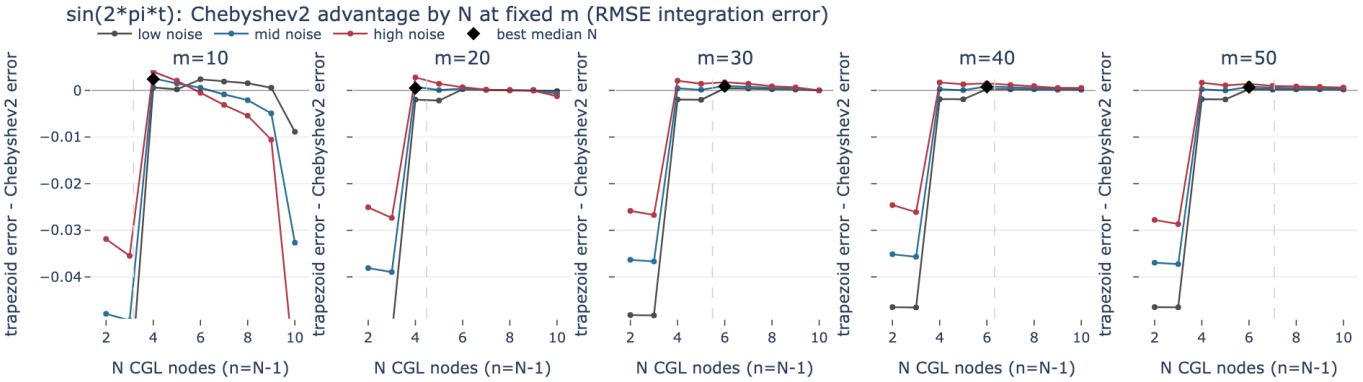


Figure 7: RMSE advantage curves for `sin(2*pi*t)` with No Tikhonov (`lambda1 = 0`).

sin(2*pi*t): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	4	4	4	75%	100%	100%	0.002417	..@@@%%=
20	4.47	3	4	6	4	0%	75%	100%	0.0005105	..@@@@@@
30	5.48	10	6	6	6	100%	100%	100%	0.0008826	..@@@@@@
40	6.32	2	6	6	6	100%	100%	100%	0.000757	..@@@@@@
50	7.07	2	6	6	6	75%	100%	100%	0.0006953	..@@@@@@

Figure 8: Robust `N` table for `sin(2*pi*t)` with No Tikhonov (`lambda1 = 0`).

3.1.3 `pi*t^(31/4)`

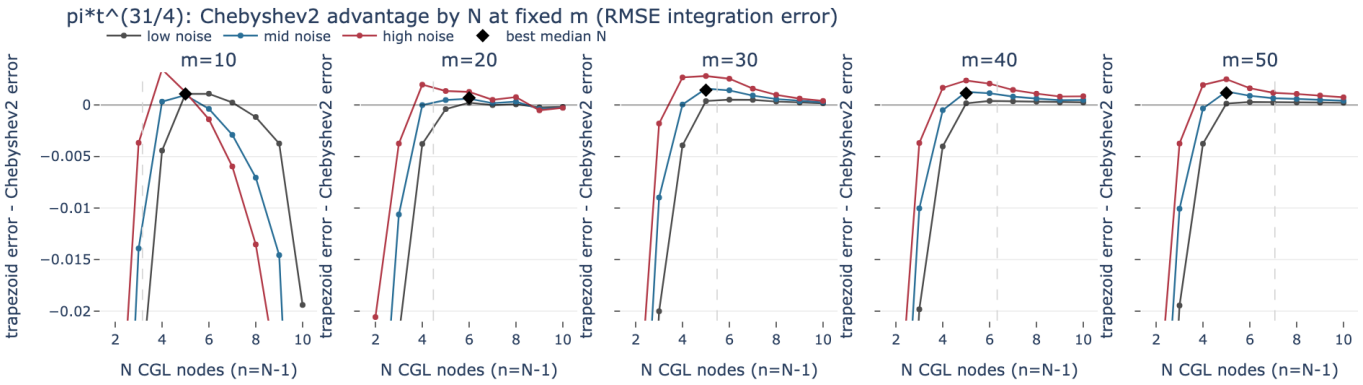


Figure 9: RMSE advantage curves for `pi*t^(31/4)` with No Tikhonov (`lambda1 = 0`).

pi*t^(31/4): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	5	5	5	100%	100%	100%	0.001068	.#@@@@%#_
20	4.47	4	6	5	5	0%	100%	100%	0.0003733	.#@@@@@@@
30	5.48	5	5	5	5	75%	100%	100%	0.001437	.#@@@@@@@
40	6.32	6	5	6	6	75%	100%	100%	0.001035	.#@@@@@@@
50	7.07	9	5	5	5	50%	100%	100%	0.001168	.#@@@@@@@

Figure 10: Robust `N` table for `pi*t^(31/4)` with No Tikhonov (`lambda1 = 0`).

3.2 Tikhonov (`lambda1 = 0.005`)

3.2.1 `tanh(t)`

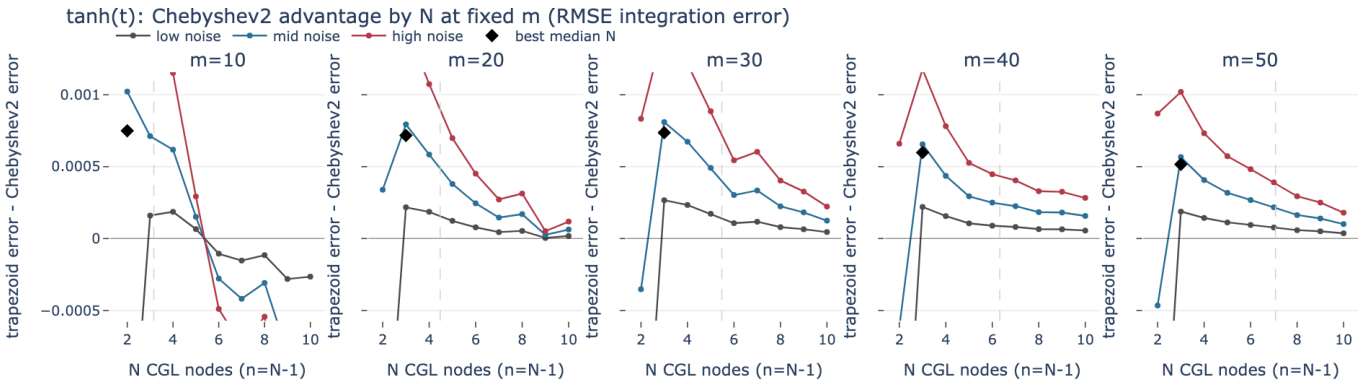


Figure 11: RMSE advantage curves for `tanh(t)` with Tikhonov (`lambda1 = 0.005`).

tanh(t): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	2	3	2	0%	75%	100%	0.0007497	@%%+−:−..
20	4.47	6	3	3	5	75%	100%	100%	0.0003448	_@#=−_:..
30	5.48	9	3	3	3	75%	100%	100%	0.0007368	.@%#++++
40	6.32	9	3	3	3	75%	100%	100%	0.000598	.@%#****
50	7.07	7	3	3	3	75%	100%	100%	0.0005158	.@%#****

Figure 12: Robust `N` table for `tanh(t)` with Tikhonov (`lambda1 = 0.005`).

3.2.2 `sin(2*pi*t)`

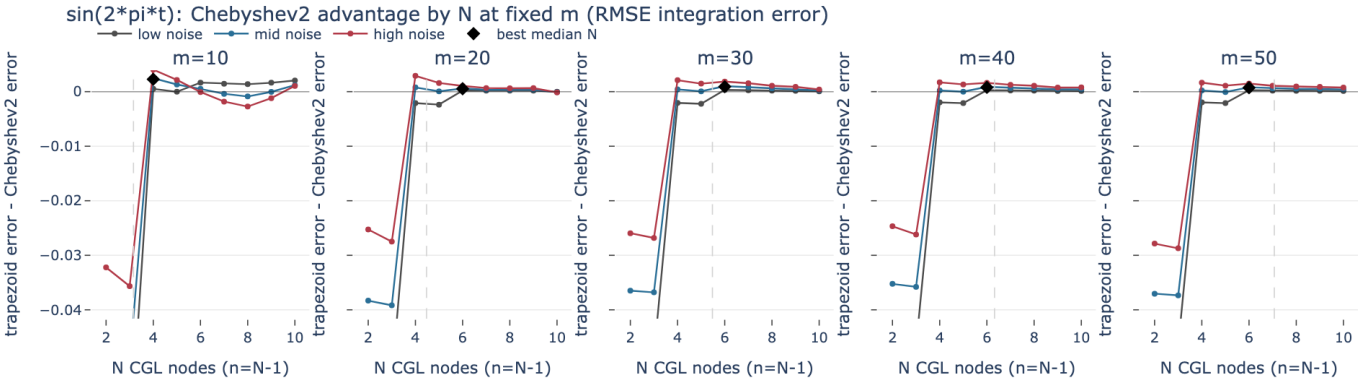


Figure 13: RMSE advantage curves for `sin(2*pi*t)` with Tikhonov (`lambda1 = 0.005`).

sin(2*pi*t): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	4	4	4	75%	100%	100%	0.002278	..@@@@@@@
20	4.47	4	6	6	6	100%	100%	100%	0.0005302	..@@@@@@@
30	5.48	10	6	6	6	75%	100%	100%	0.0009199	..@@@@@@@
40	6.32	2	6	6	6	75%	100%	100%	0.0008021	..@@@@@@@
50	7.07	2	6	6	6	75%	100%	100%	0.0007325	..@@@@@@@

Figure 14: Robust `N` table for `sin(2*pi*t)` with Tikhonov (`lambda1 = 0.005`).

3.2.3 `pi*t^(31/4)`

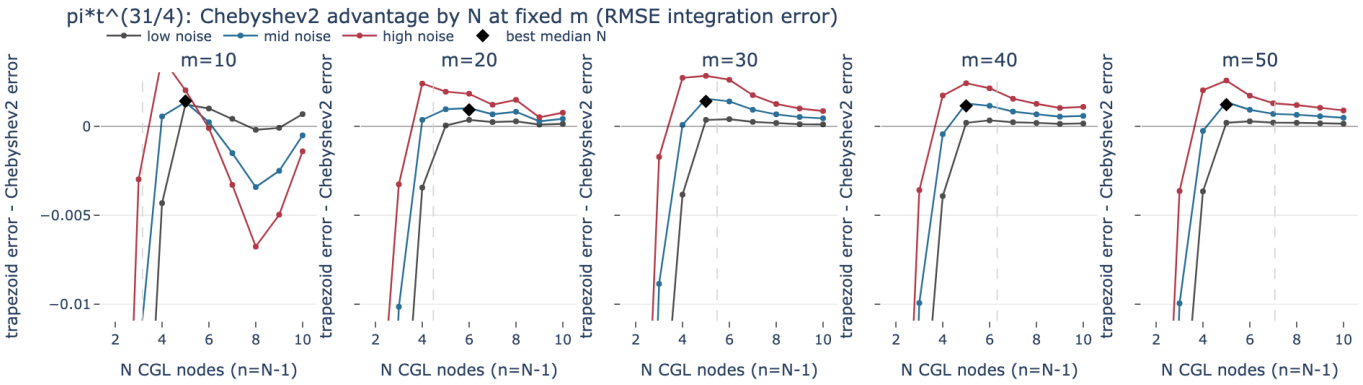


Figure 15: RMSE advantage curves for `pi*t^(31/4)` with Tikhonov (`lambda1 = 0.005`).

pi*t^(31/4): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	6	5	5	5	100%	100%	100%	0.00142	.#@@@@@@
20	4.47	3	6	5	6	75%	100%	100%	0.00093	.#@@@@@@@
30	5.48	6	5	5	5	75%	100%	100%	0.001406	.#@@@@@@@
40	6.32	5	5	6	5	50%	100%	100%	0.001152	.#@@@@@@@
50	7.07	10	5	5	5	50%	100%	100%	0.001218	.#@@@@@@@

Figure 16: Robust `N` table for `pi*t^(31/4)` with Tikhonov (`lambda1 = 0.005`).

4 Random Degree-4 Polynomials

The random-polynomial notebook broadens the scalar comparison beyond the three hand-picked analytic functions. Each degree-4 polynomial is defined by fixed random values at five CGL nodes on `[0, 1]`; the same Monte Carlo grid is then used for no Tikhonov and for `lambda1 = 0.005`.

Function	f(0.000)	f(0.146)	f(0.500)	f(0.854)	f(1.000)
random degree-4 polynomial 1	-0.1211	0.3145	0.1258	0.4067	-0.3488
random degree-4 polynomial 2	-0.1641	-0.0227	0.0029	-0.1796	0.199
random degree-4 polynomial 3	-0.4065	0.4601	-0.2177	0.1236	0.423

4.1 No Tikhonov (`lambda1 = 0`)

4.1.1 `random degree-4 polynomial 1`

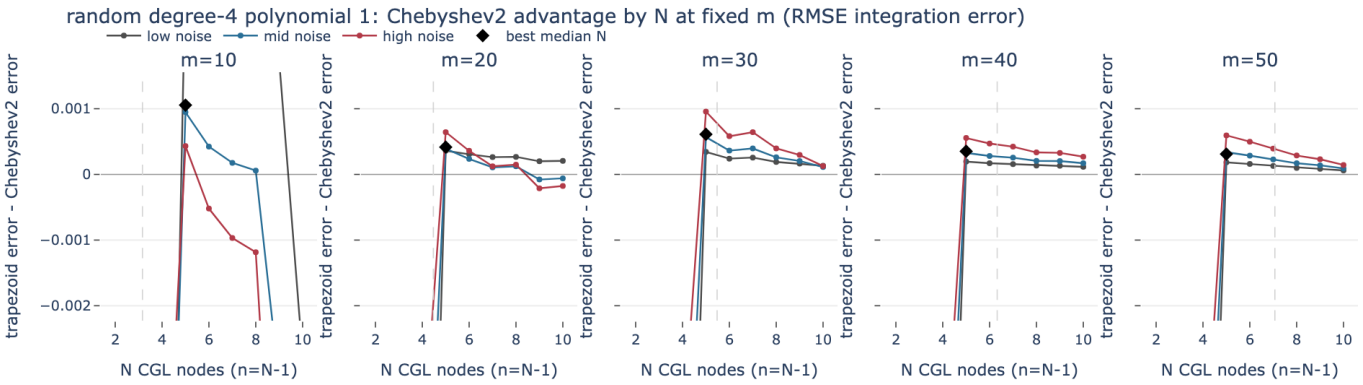


Figure 17: RMSE advantage curves for `random degree-4 polynomial 1` with No Tikhonov (`lambda1 = 0`).

random degree-4 polynomial 1: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	6	5	5	5	100%	100%	100%	0.001055	.+=@@@@#:
20	4.47	6	5	5	5	100%	100%	100%	0.0004163	._:@@@@@@
30	5.48	9	5	5	5	100%	100%	100%	0.0006104	.__@@@@@@
40	6.32	9	5	5	5	100%	100%	100%	0.0003521	.._@@@@@@
50	7.07	8	5	5	5	100%	100%	100%	0.0003103	.._@@@@@@

Figure 18: Robust `N` table for `random degree-4 polynomial 1` with No Tikhonov (`lambda1 = 0`).

4.1.2 `random degree-4 polynomial 2`

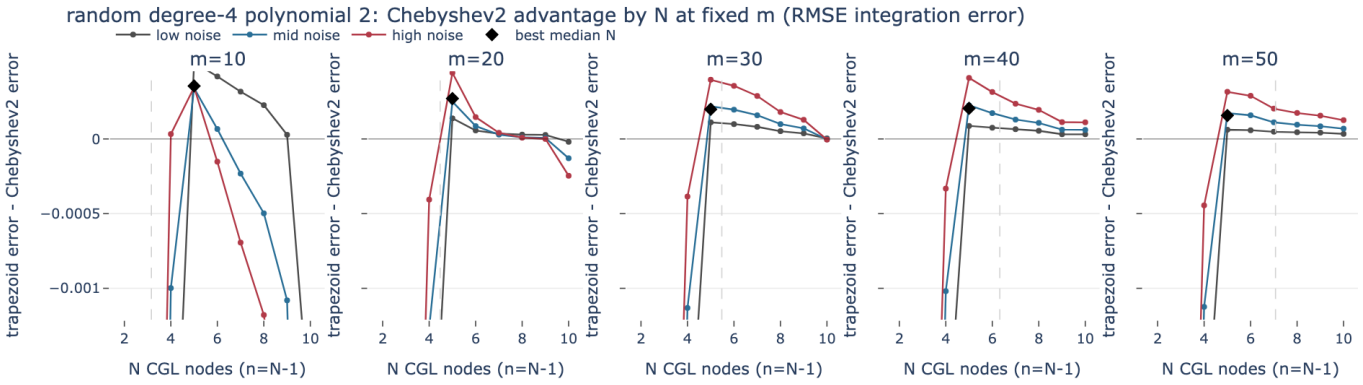


Figure 19: RMSE advantage curves for `random degree-4 polynomial 2` with No Tikhonov (`lambda1 = 0`).

random degree-4 polynomial 2: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	5	5	5	100%	100%	100%	0.000355	..%@@@%%=
20	4.47	6	5	5	5	100%	100%	100%	0.0002709	..#@@@@@@
30	5.48	10	5	5	5	100%	100%	100%	0.0001986	..#@@@@@@
40	6.32	6	5	5	5	100%	100%	100%	0.0002049	..%@@@@@@
50	7.07	9	5	5	5	100%	100%	100%	0.0001571	..#@@@@@@

Figure 20: Robust `N` table for `random degree-4 polynomial 2` with No Tikhonov (`lambda1 = 0`).

4.1.3 `random degree-4 polynomial 3`

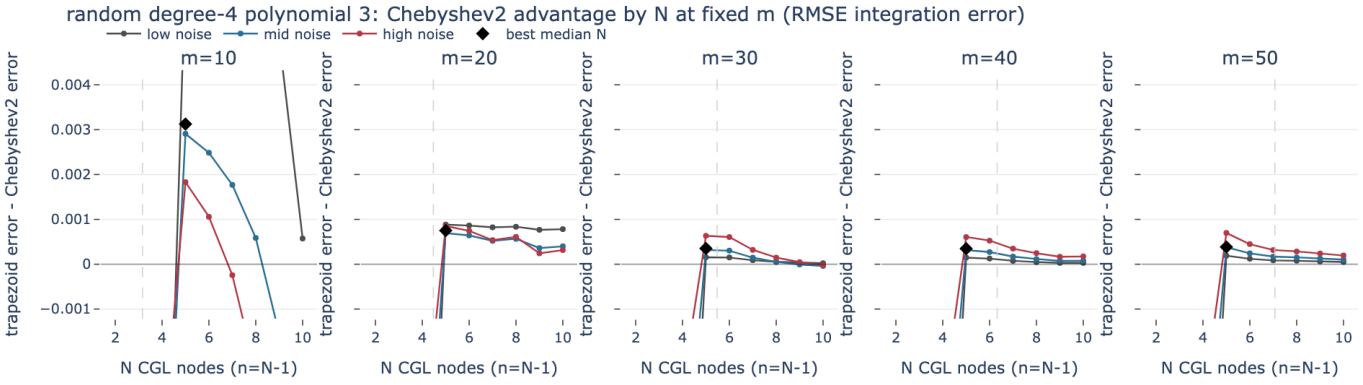


Figure 21: RMSE advantage curves for `random degree-4 polynomial 3` with No Tikhonov (`lambda1 = 0`).

random degree-4 polynomial 3: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	5	5	5	100%	100%	100%	0.003125	..#@@@%+
20	4.47	5	5	5	5	100%	100%	100%	0.0007515	._#@@@@@
30	5.48	2	5	5	5	100%	100%	100%	0.0003505	.:#@@@@@
40	6.32	2	5	5	5	100%	100%	100%	0.0003465	.-#@@@@@
50	7.07	9	5	5	5	100%	100%	100%	0.000384	.-#@@@@@

Figure 22: Robust `N` table for `random degree-4 polynomial 3` with No Tikhonov (`lambda1 = 0`).

4.2 Tikhonov (`lambda1 = 0.005`)

4.2.1 `random degree-4 polynomial 1`

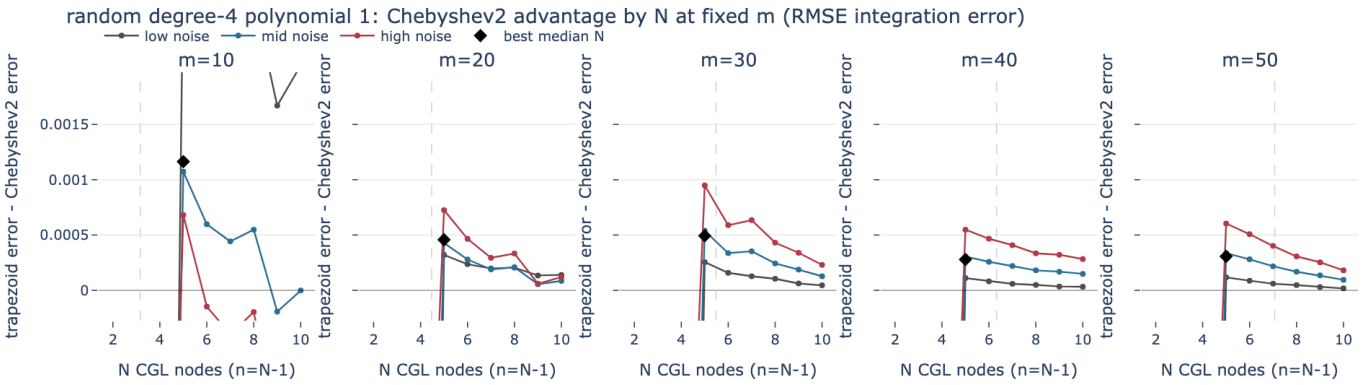


Figure 23: RMSE advantage curves for `random degree-4 polynomial 1` with Tikhonov (`lambda1 = 0.005`).

random degree-4 polynomial 1: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	5	5	5	100%	100%	100%	0.001163	.+=@@@*@
20	4.47	5	5	5	5	100%	100%	100%	0.0004577	._:@@@@@@
30	5.48	7	5	5	7	100%	100%	100%	0.0003219	.__@@@@@@
40	6.32	7	5	5	7	75%	100%	100%	0.0001983	.._@@@@@@
50	7.07	7	5	5	5	75%	100%	100%	0.0003066	.._@@@@@@

Figure 24: Robust `N` table for `random degree-4 polynomial 1` with Tikhonov (`lambda1 = 0.005`).

4.2.2 `random degree-4 polynomial 2`

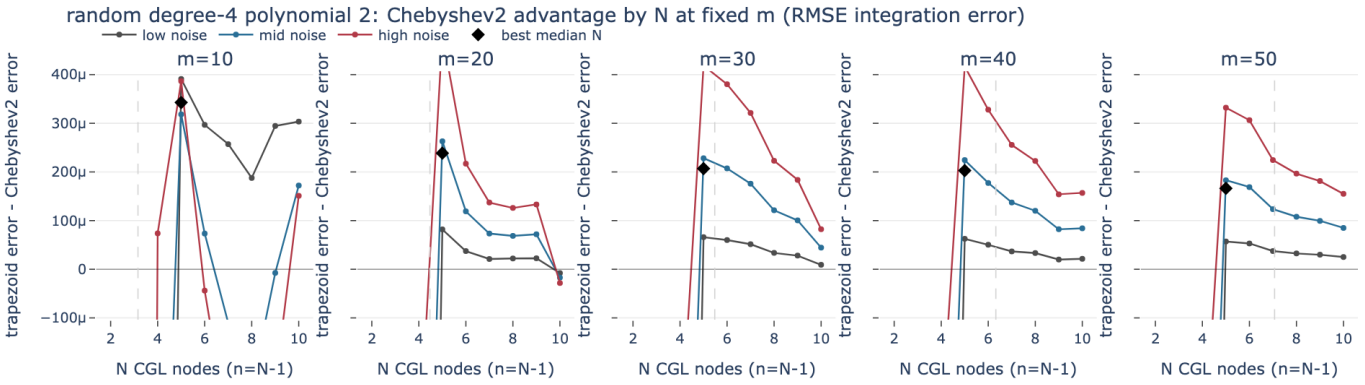


Figure 25: RMSE advantage curves for `random degree-4 polynomial 2` with Tikhonov (`lambda1 = 0.005`).

random degree-4 polynomial 2: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	10	5	5	5	100%	100%	100%	0.0003427	..%@@@@@@
20	4.47	5	5	5	5	75%	100%	100%	0.0002387	..#@@@@@@
30	5.48	7	5	5	5	75%	100%	100%	0.000207	..#@@@@@@
40	6.32	5	5	5	5	75%	100%	100%	0.0002027	..#@@@@@@
50	7.07	9	5	5	5	75%	100%	100%	0.0001663	..#@@@@@@

Figure 26: Robust `N` table for `random degree-4 polynomial 2` with Tikhonov (`lambda1 = 0.005`).

4.2.3 `random degree-4 polynomial 3`

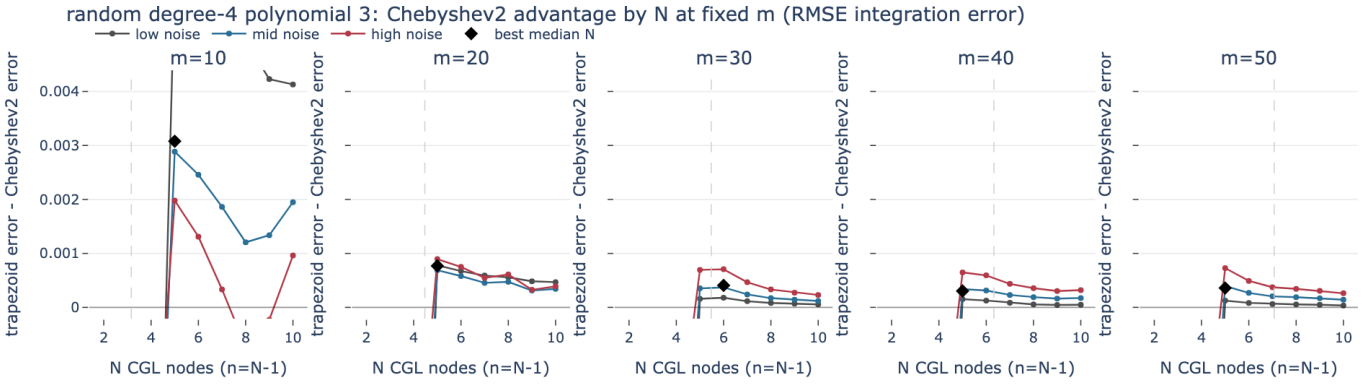


Figure 27: RMSE advantage curves for `random degree-4 polynomial 3` with Tikhonov (`lambda1 = 0.005`).

random degree-4 polynomial 3: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	6	5	5	5	100%	100%	100%	0.003078	..#@@@@@@
20	4.47	5	5	5	5	100%	100%	100%	0.0007689	._#@@@@@@
30	5.48	2	6	5	6	100%	100%	100%	0.0004067	.:#@@@@@@
40	6.32	2	5	5	5	100%	100%	100%	0.0003033	.-#@@@@@@
50	7.07	9	5	5	5	100%	100%	100%	0.000358	.-#@@@@@@

Figure 28: Robust `N` table for `random degree-4 polynomial 3` with Tikhonov (`lambda1 = 0.005`).

5 EuRoC Aggressive 200 ms Snippets

The EuRoC extension changes the question from synthetic scalar functions to IMU-like signals. For each merged EuRoC CSV, the workflow fits one-second `gyro_norm` windows with Chebyshev2 using `N = 50`. It then scores only the center interval `[0.4, 0.6]` with the local aggressive-window diagnostic, so the selected window is aggressive in the 200 ms interval actually used by the quadrature comparison. The resulting CGL node values define the reference scalar function; the Monte Carlo comparison samples only that middle 200 ms interval.

The figure below shows all selected snippets. Points are the raw uniformly sampled IMU values in the selected one-second window, the line is the `N = 50` Chebyshev2 reference fit, and the shaded region is the 200 ms interval used for quadrature.

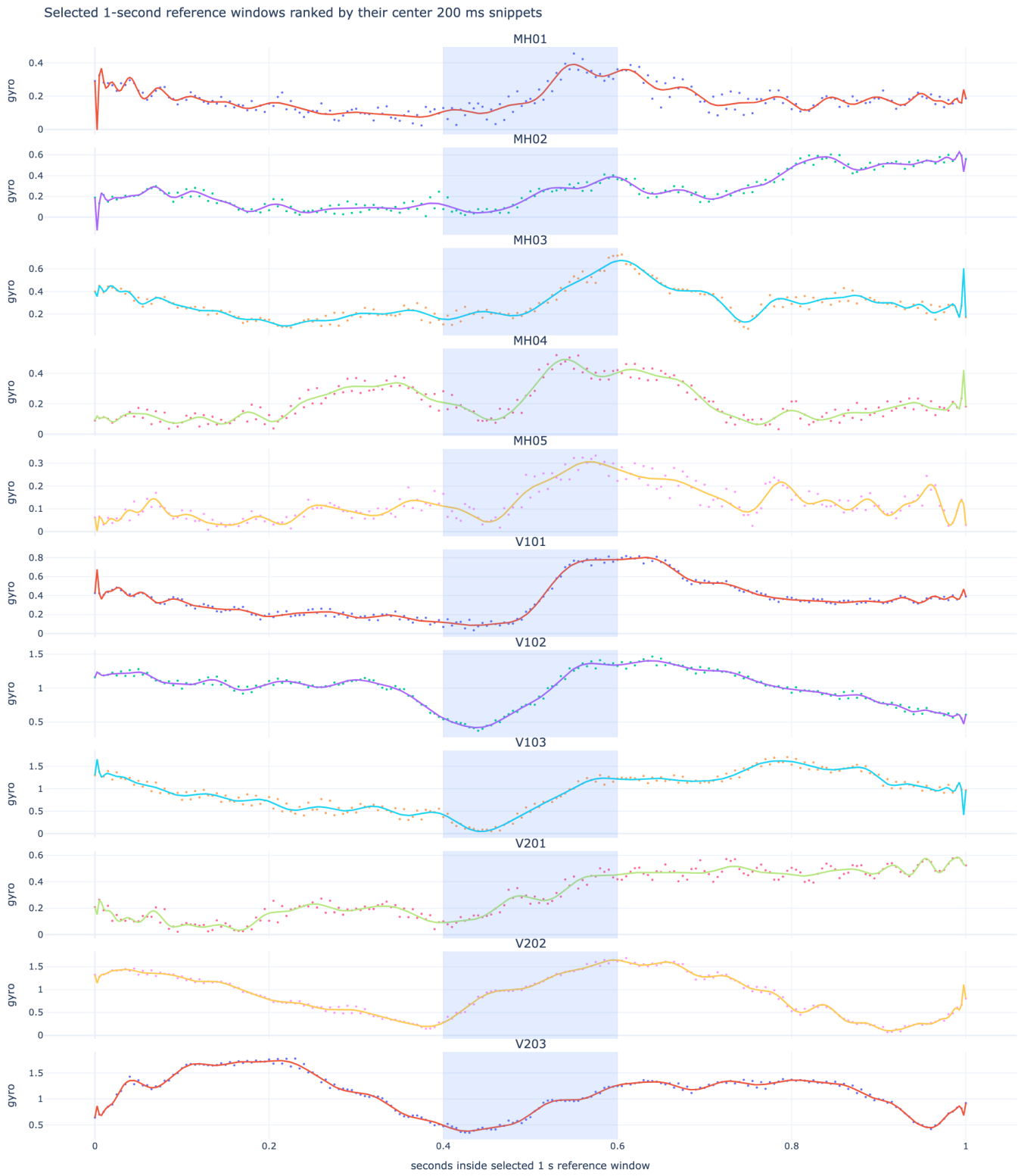


Figure 29: All selected EuRoC aggressive snippets.

Selection metadata:

dataset	window	snippet_s	score	activity	high_ratio	nRMSE
V101	23	23.4-23.6	13.87	1.621	0.8876	0.1391
V103	68	68.4-68.6	9.266	0.8771	0.8588	0.1117

dataset	window	snippet_s	score	activity	high_ratio	nRMSE
V102	69	69.4-69.6	8.143	0.929	0.7507	0.07532
MH02	53	53.4-53.6	7.924	1.2	0.8361	0.2954
MH03	45	45.4-45.6	7.782	0.9836	0.7857	0.1817
MH04	49	49.4-49.6	6.888	1.038	0.7596	0.2433
V202	54	54.4-54.6	6.555	0.9056	0.7245	0.06962
V201	85	85.4-85.6	6.465	0.7833	0.7612	0.1882
MH01	38	38.4-38.6	6.373	1.017	0.7885	0.3344
MH05	16	16.4-16.6	6.302	0.9104	0.7986	0.2625
V203	73	73.4-73.6	5.435	0.6895	0.7481	0.06814

6 EuRoC Decision-oriented Views

The following pages show the same decision views for every selected EuRoC snippet, first without Tikhonov and then with `lambda1 = 0.005`.

6.1 No Tikhonov (`lambda1 = 0`)

6.1.1 `MH01 center-200ms aggressive gyro_norm`

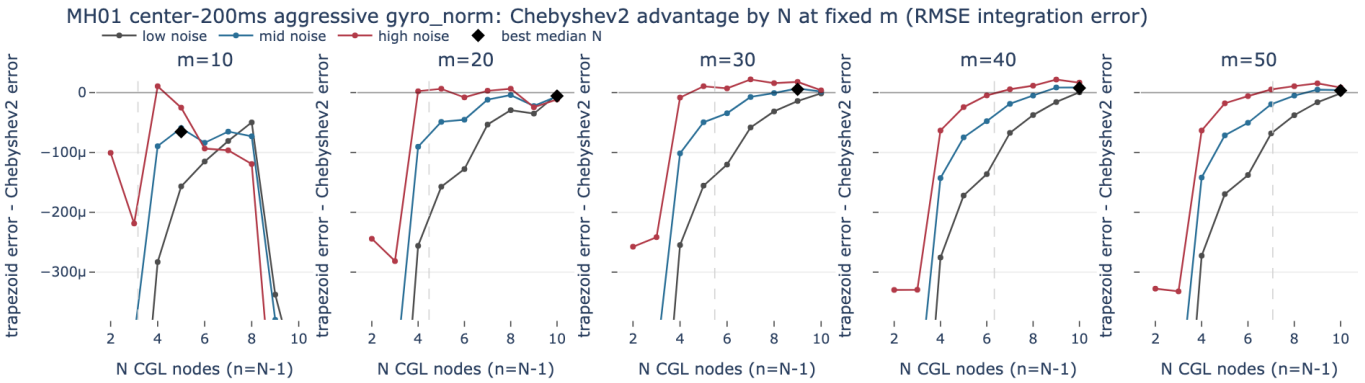


Figure 30: RMSE advantage curves for `MH01 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

MH01 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	5	7	4	0%	0%	67%	-0.0001047	**@@@@*.
20	4.47	2	10	10	8	0%	25%	100%	-5.878e-06	._#%%@@@@
30	5.48	10	9	10	10	25%	100%	100%	1.379e-06	._#%%@@@@
40	6.32	9	10	10	10	50%	100%	100%	7.669e-06	..#%%@@@@
50	7.07	2	10	10	10	50%	100%	100%	3.415e-06	._#%%@@@@

Figure 31: Robust `N` table for `MH01 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.2 MH02 center-200ms aggressive gyro_norm

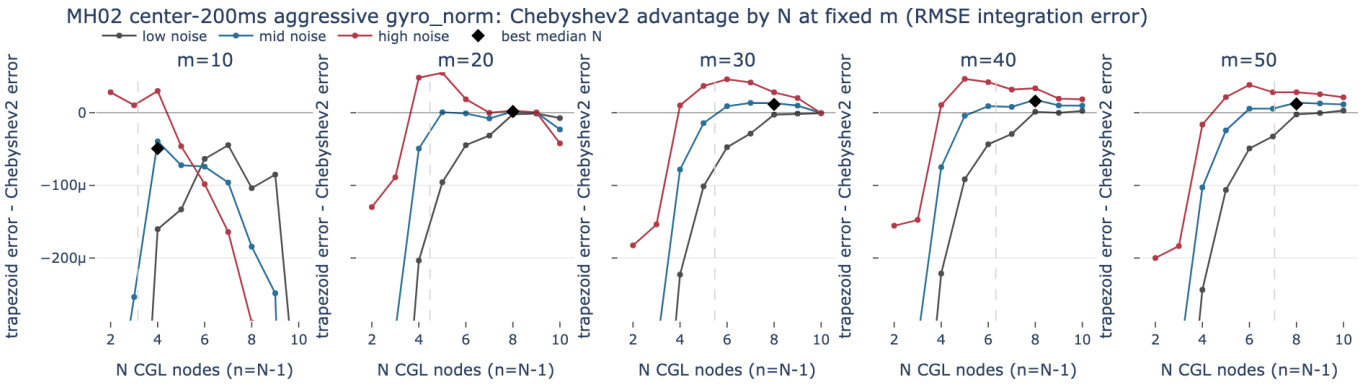


Figure 32: RMSE advantage curves for MH02 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

MH02 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	4	5	4	0%	0%	100%	-4.938e-05	+#@@@@%#.
20	4.47	5	8	8	8	25%	100%	100%	1.559e-06	.-%@@@@@@
30	5.48	10	8	8	8	50%	100%	100%	1.142e-05	.:#%@@@@@
40	6.32	4	8	8	8	50%	100%	100%	1.593e-05	.:#%@@@@@
50	7.07	4	8	10	10	75%	100%	100%	1.026e-05	.:#%@@@@@

Figure 33: Robust N table for MH02 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

6.1.3 MH03 center-200ms aggressive gyro_norm

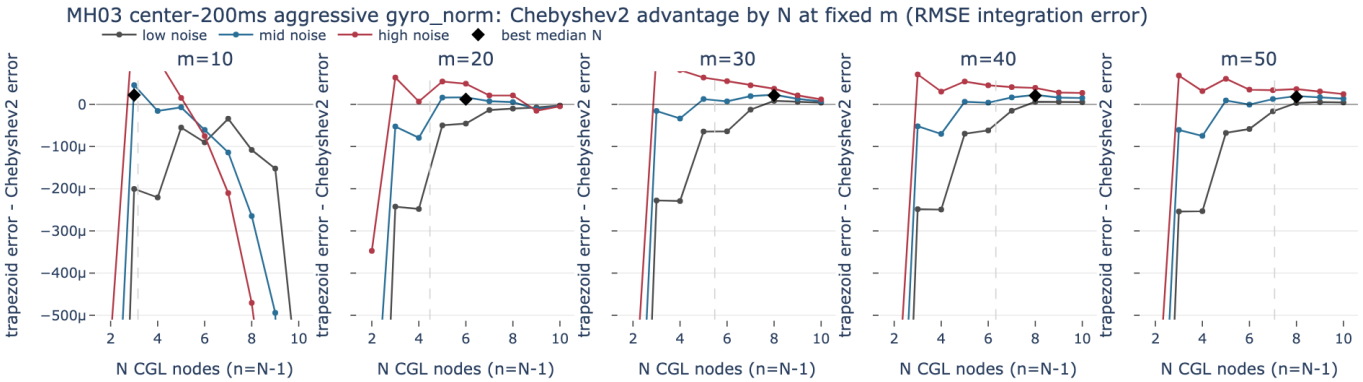


Figure 34: RMSE advantage curves for MH03 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

MH03 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	3	3	3	0%	75%	100%	2.183e-05	-@@@%%#.
20	4.47	2	6	5	5	0%	75%	100%	1.098e-05	.%%@@@@@@
30	5.48	4	8	8	8	75%	100%	100%	2.11e-05	.@%@@@@@@
40	6.32	4	8	8	8	75%	100%	100%	2.048e-05	.%%@@@@@@
50	7.07	9	8	8	8	50%	100%	100%	1.77e-05	.%%@@@@@@

Figure 35: Robust `N` table for `MH03 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.4 `MH04 center-200ms aggressive gyro_norm`

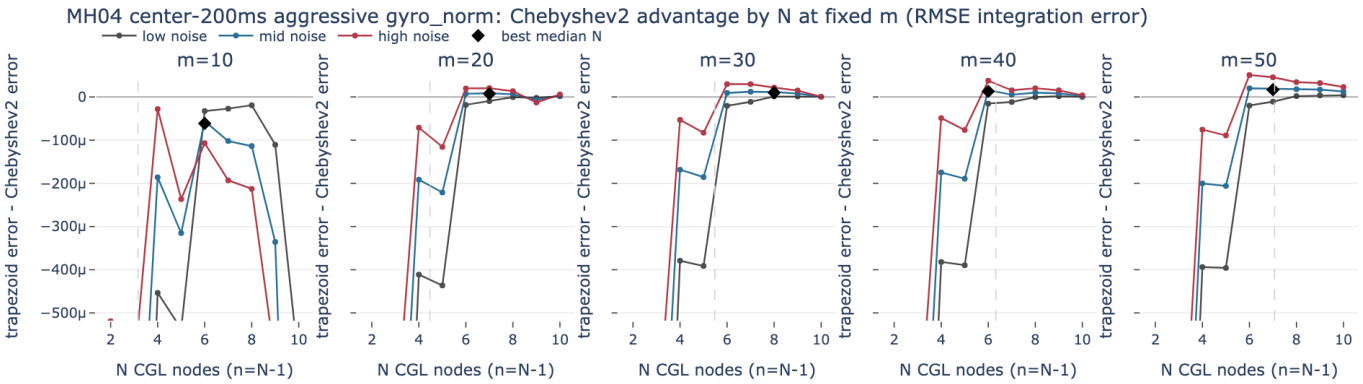


Figure 36: RMSE advantage curves for `MH04 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

MH04 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	6	6	6	0%	0%	0%	-6.136e-05	--%#@@@%.
20	4.47	2	7	6	7	0%	100%	100%	7.339e-06	..##@@@@@
30	5.48	2	8	8	8	50%	100%	100%	1.027e-05	..%#@@@@@
40	6.32	2	6	6	6	0%	100%	100%	1.295e-05	..%#@@@@@
50	7.07	2	7	6	8	50%	100%	100%	1.614e-05	..##@@@@@

Figure 37: Robust `N` table for `MH04 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.5 `MH05 center-200ms aggressive gyro_norm`

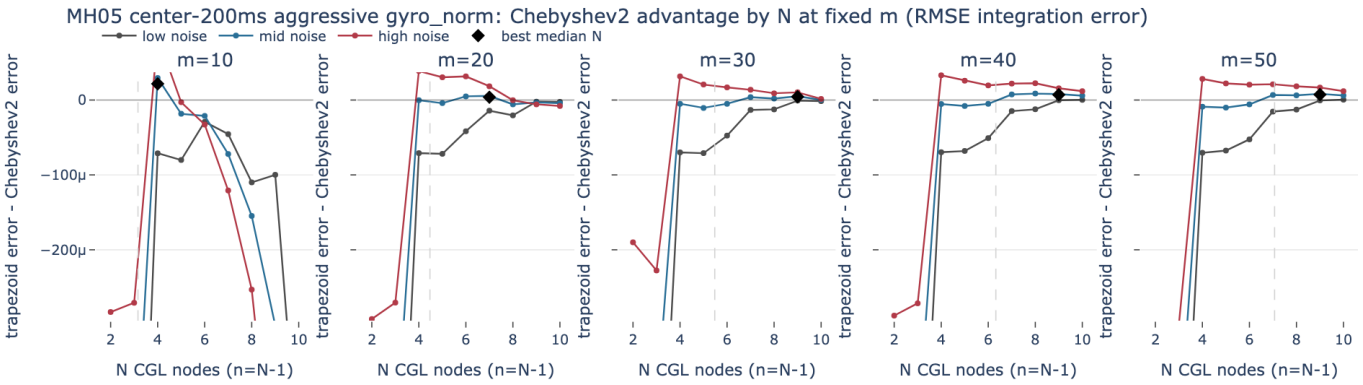


Figure 38: RMSE advantage curves for `MH05 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

MH05 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	4	4	4	0%	75%	100%	2.16e-05	=+@@@%#.
20	4.47	5	7	4	4	0%	50%	100%	-5.794e-06	._@@@@@@@
30	5.48	2	9	9	8	0%	75%	100%	9.685e-07	..@@@@@@@
40	6.32	8	9	9	8	0%	100%	100%	6.811e-06	._@@@@@@@
50	7.07	2	9	9	9	50%	100%	100%	7.325e-06	._@@@@@@@

Figure 39: Robust `N` table for `MH05 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.6 `V101 center-200ms aggressive gyro_norm`

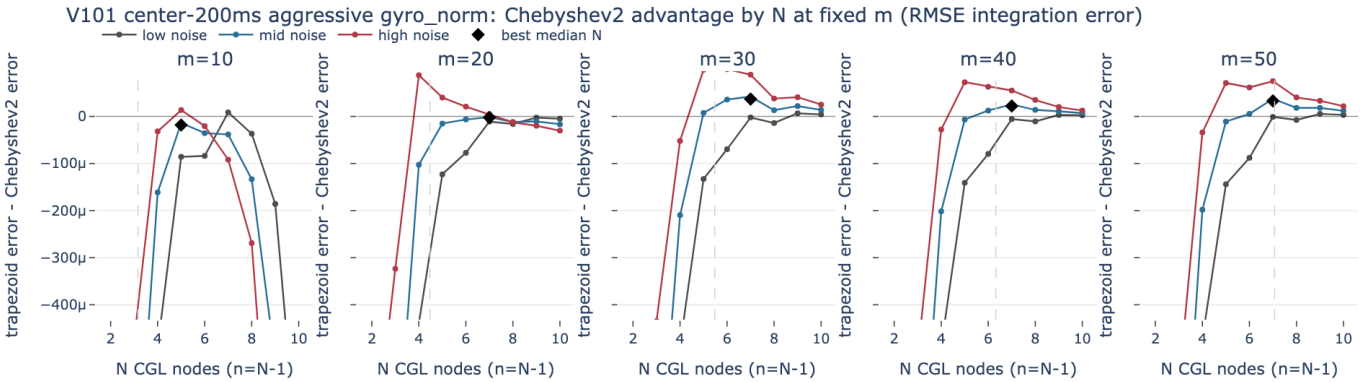


Figure 40: RMSE advantage curves for `V101 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

V101 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	5	6	5	0%	25%	100%	-1.834e-05	_+%@@@@#.
20	4.47	2	7	7	7	0%	25%	100%	-2.433e-06	.=%@@@@@@
30	5.48	6	7	7	6	0%	100%	100%	2.71e-05	.=%@@@@@@
40	6.32	2	7	7	7	50%	100%	100%	2.201e-05	.=%@@@@@@
50	7.07	2	7	7	7	50%	100%	100%	3.288e-05	.=%@@@@@@

Figure 41: Robust `N` table for `V101 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.7 `V102 center-200ms aggressive gyro_norm`

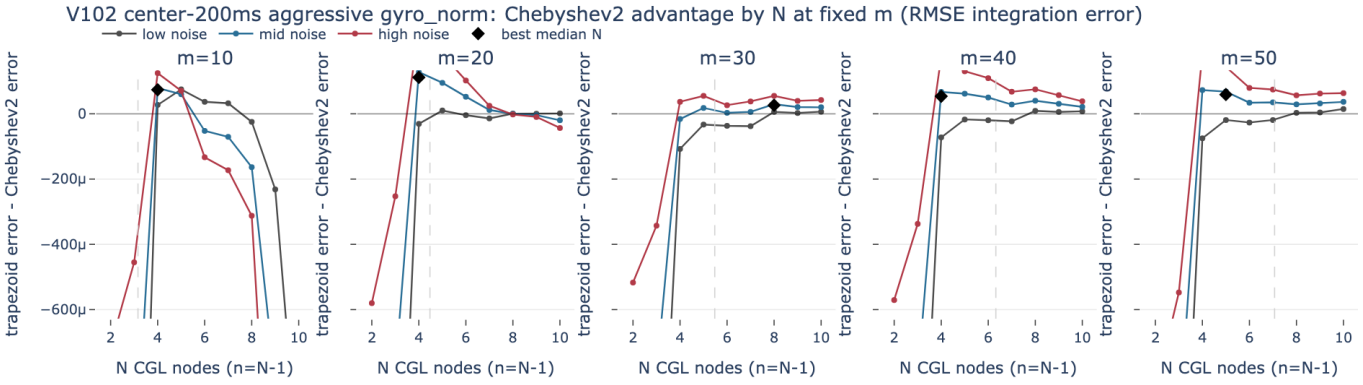


Figure 42: RMSE advantage curves for `V102 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

V102 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	7	4	5	5	100%	100%	100%	6.363e-05	-+@@@@%#.
20	4.47	4	4	4	4	50%	100%	100%	0.0001115	.-@@@%%%%
30	5.48	7	8	5	8	75%	100%	100%	2.642e-05	.-@@@@@@@
40	6.32	3	4	4	4	0%	100%	100%	5.37e-05	.-@@@@@@@
50	7.07	4	5	5	4	0%	100%	100%	5.839e-05	.-@@@@@@@

Figure 43: Robust `N` table for `V102 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.8 `V103 center-200ms aggressive gyro_norm`

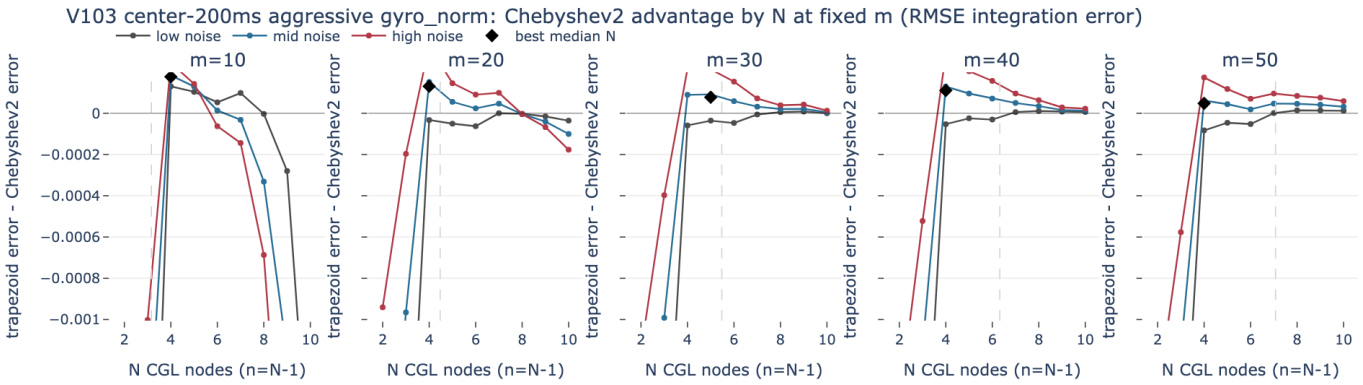


Figure 44: RMSE advantage curves for `V103 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

V103 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	6	4	4	4	100%	100%	100%	0.0001765	:*@@@@%#.
20	4.47	4	4	4	4	50%	100%	100%	0.0001307	.+@@@@@%%
30	5.48	5	5	4	5	25%	100%	100%	7.73e-05	.+@@@@@@@
40	6.32	3	4	4	4	25%	100%	100%	0.0001101	.=@@@@@@@
50	7.07	7	4	4	7	50%	100%	100%	4.14e-05	.=@@@@@@@

Figure 45: Robust `N` table for `V103 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.9 `V201 center-200ms aggressive gyro_norm`

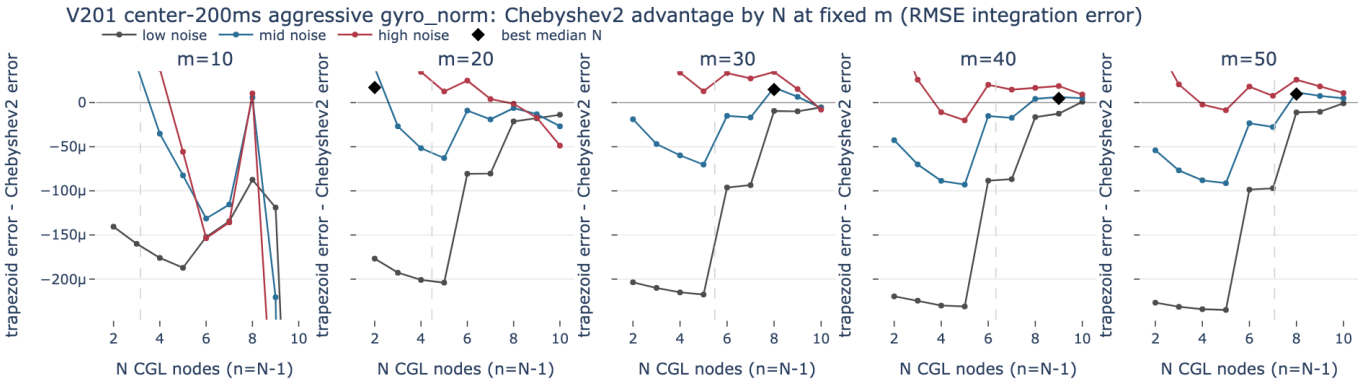


Figure 46: RMSE advantage curves for `V201 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

V201 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	2	2	2	0%	100%	100%	9.226e-05	@@%%%@#.
20	4.47	2	2	2	2	0%	75%	100%	1.711e-05	@-_.*+#*+
30	5.48	5	8	8	8	25%	100%	100%	1.485e-05	=:_.**@%#
40	6.32	9	9	10	9	0%	100%	100%	4.744e-06	=:..##@@@
50	7.07	6	8	8	8	25%	100%	100%	9.781e-06	-_..**@@@

Figure 47: Robust `N` table for `V201 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.10 `V202 center-200ms aggressive gyro_norm`

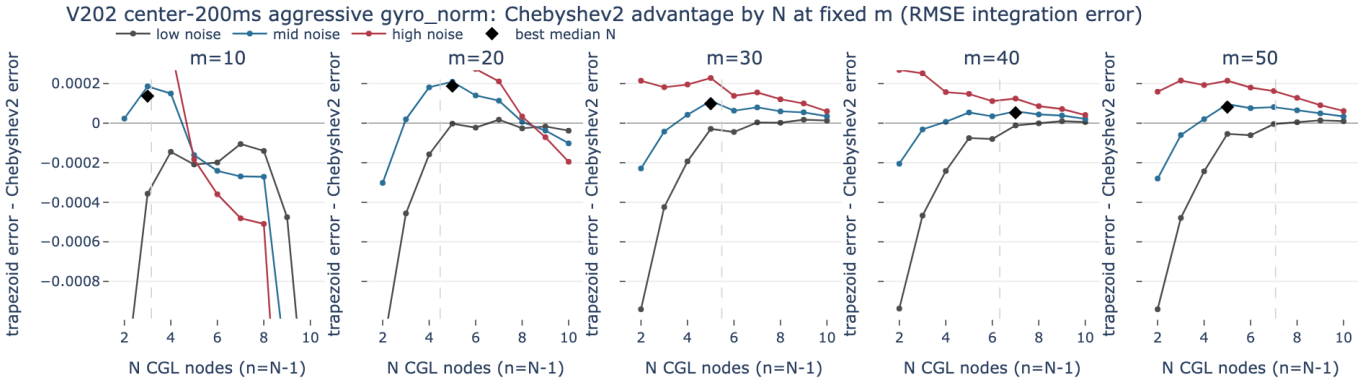


Figure 48: RMSE advantage curves for `V202 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

V202 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	3	3	3	0%	75%	100%	0.0001369	@@@%%%#.
20	4.47	4	5	5	4	0%	100%	100%	0.0001468	.*%@%%*+
30	5.48	5	5	5	5	50%	100%	100%	9.855e-05	.+#@%%%#
40	6.32	7	7	5	7	50%	100%	100%	5.253e-05	.*#@%@@%
50	7.07	7	5	5	5	25%	100%	100%	8.105e-05	.+#@@@%%

Figure 49: Robust `N` table for `V202 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.11 `V203 center-200ms aggressive gyro_norm`

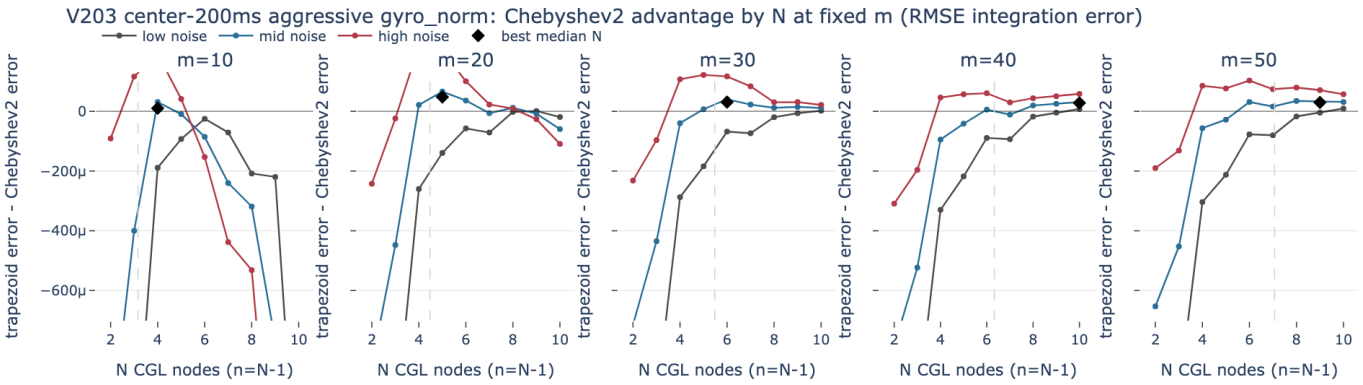


Figure 50: RMSE advantage curves for `V203 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

V203 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	4	5	4	0%	75%	100%	1.02e-05	*%@@@%#.
20	4.47	6	5	5	5	0%	75%	100%	4.799e-05	.=@@@%%%
30	5.48	6	6	6	6	0%	100%	100%	3.06e-05	.-%@@@@@
40	6.32	6	10	8	10	75%	100%	100%	2.798e-05	.-%%@@@@
50	7.07	7	9	8	8	25%	100%	100%	2.922e-05	.-%%@@@@

Figure 51: Robust `N` table for `V203 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.2 Tikhonov (`lambda1 = 0.005`)

6.2.1 `MH01 center-200ms aggressive gyro_norm`

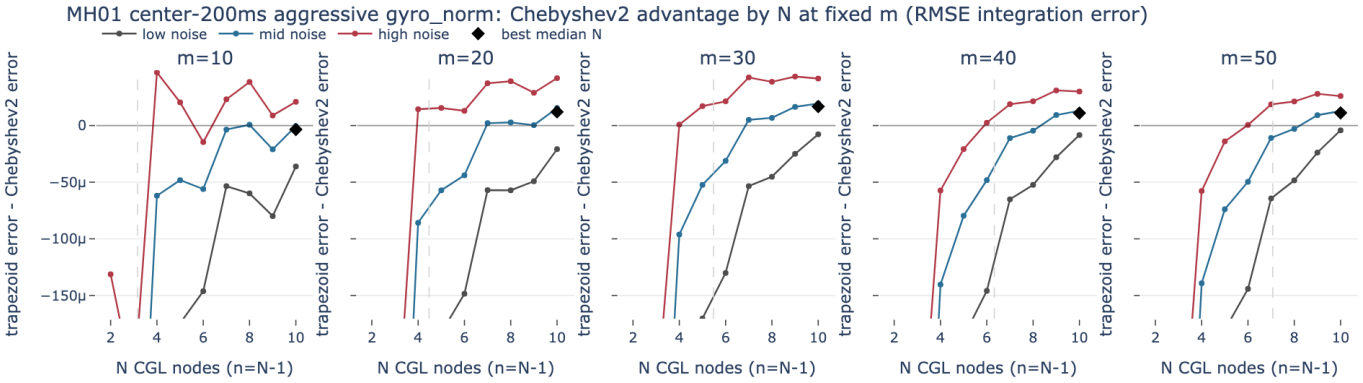


Figure 52: RMSE advantage curves for `MH01 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

MH01 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	10	10	8	0%	50%	100%	−4.441e−06	..%%%@@@@
20	4.47	2	10	10	10	0%	100%	100%	1.205e−05	._#%%@@@@
30	5.48	9	10	10	10	50%	100%	100%	1.678e−05	._#%%@@@@
40	6.32	9	10	10	9	0%	75%	100%	6.305e−06	..#%%@@@@
50	7.07	2	10	10	10	50%	100%	100%	1.117e−05	._#%%@@@@

Figure 53: Robust `N` table for `MH01 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.2 `MH02 center-200ms aggressive gyro_norm`

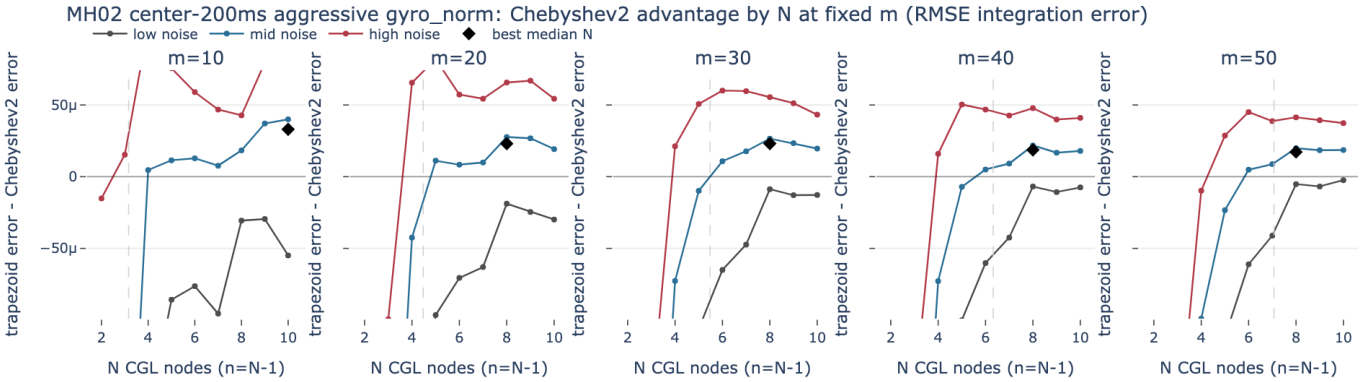


Figure 54: RMSE advantage curves for `MH02 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

MH02 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	9	10	10	10	0%	100%	100%	3.302e−05	.=%@@@@@
20	4.47	6	8	8	8	25%	100%	100%	2.3e−05	.-%@@@@@
30	5.48	7	8	8	8	50%	100%	100%	2.306e−05	.:#%@@@@
40	6.32	6	8	8	8	50%	100%	100%	1.869e−05	.:#%@@@@
50	7.07	4	8	10	10	50%	100%	100%	1.637e−05	.:#%@@@@

Figure 55: Robust `N` table for `MH02 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.3 MH03 center-200ms aggressive gyro_norm

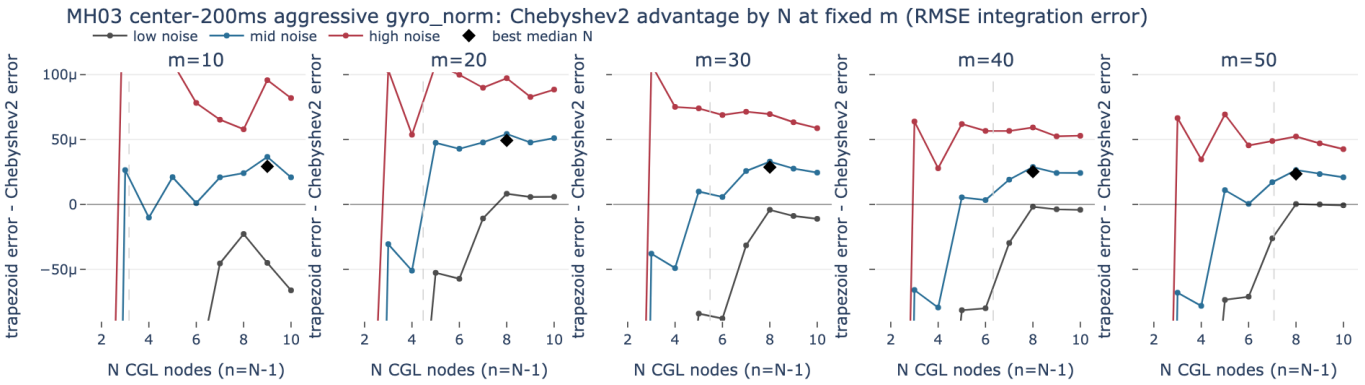


Figure 56: RMSE advantage curves for MH03 center-200ms aggressive gyro_norm with Tikhonov (lambda1 = 0.005).

MH03 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	9	5	9	0%	100%	100%	2.933e-05	.@@@@@@@
20	4.47	2	8	5	5	0%	100%	100%	3.95e-05	.%%@@@@@@
30	5.48	4	8	8	8	50%	100%	100%	2.878e-05	.%%@@@@@@
40	6.32	3	8	8	8	50%	100%	100%	2.53e-05	.%%@@@@@@
50	7.07	9	8	8	9	50%	100%	100%	2.072e-05	.%%@@@@@@

Figure 57: Robust N table for MH03 center-200ms aggressive gyro_norm with Tikhonov (lambda1 = 0.005).

6.2.4 MH04 center-200ms aggressive gyro_norm

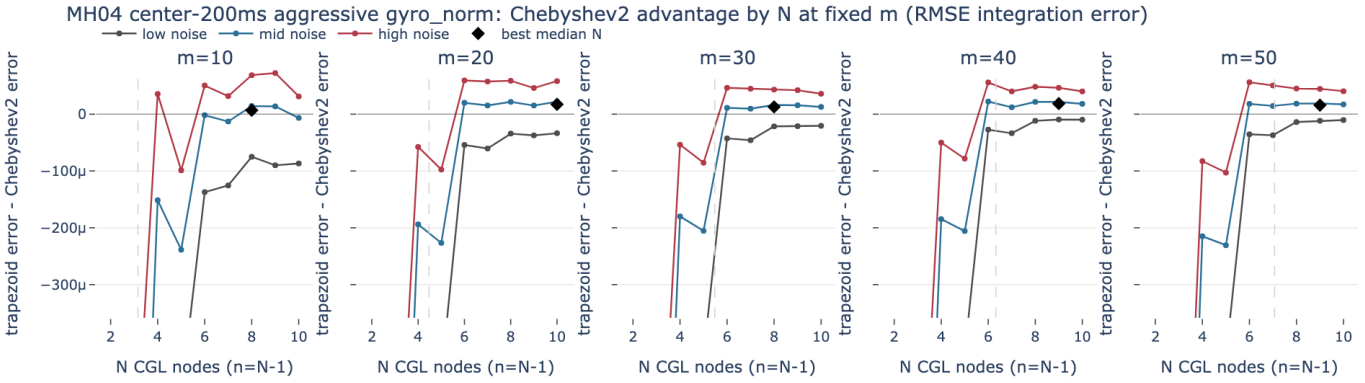


Figure 58: RMSE advantage curves for MH04 center-200ms aggressive gyro_norm with Tikhonov (lambda1 = 0.005).

MH04 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	8	6	6	0%	50%	100%	−1.005e−05	..%#@@@@@
20	4.47	2	10	6	8	0%	100%	100%	1.689e−05	..##@@@@@
30	5.48	2	8	8	8	0%	100%	100%	1.276e−05	..##@@@@@
40	6.32	2	9	8	8	25%	100%	100%	1.842e−05	..##@@@@@
50	7.07	10	9	9	9	25%	100%	100%	1.584e−05	..##@@@@@

Figure 59: Robust `N` table for `MH04 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.5 `MH05 center-200ms aggressive gyro_norm`

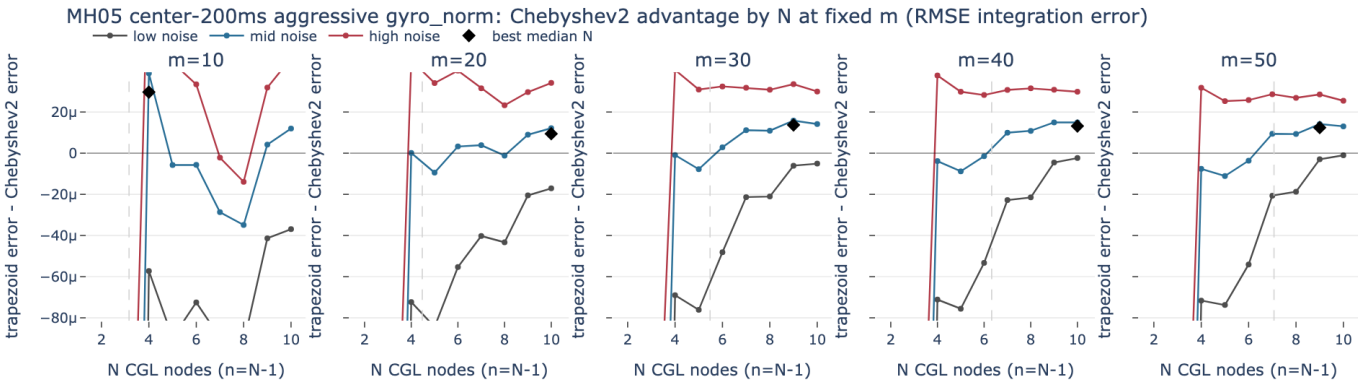


Figure 60: RMSE advantage curves for `MH05 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

MH05 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	3	4	4	4	0%	100%	100%	2.966e−05	._@%@%%@@
20	4.47	5	10	10	10	0%	100%	100%	9.424e−06	._@@@@@@@
30	5.48	2	9	10	10	50%	100%	100%	1.224e−05	..@@@@@@@
40	6.32	8	10	10	8	0%	100%	100%	8.168e−06	._@@@@@@@
50	7.07	2	9	9	9	50%	100%	100%	1.241e−05	._@@@@@@@

Figure 61: Robust `N` table for `MH05 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.6 `V101 center-200ms aggressive gyro_norm`

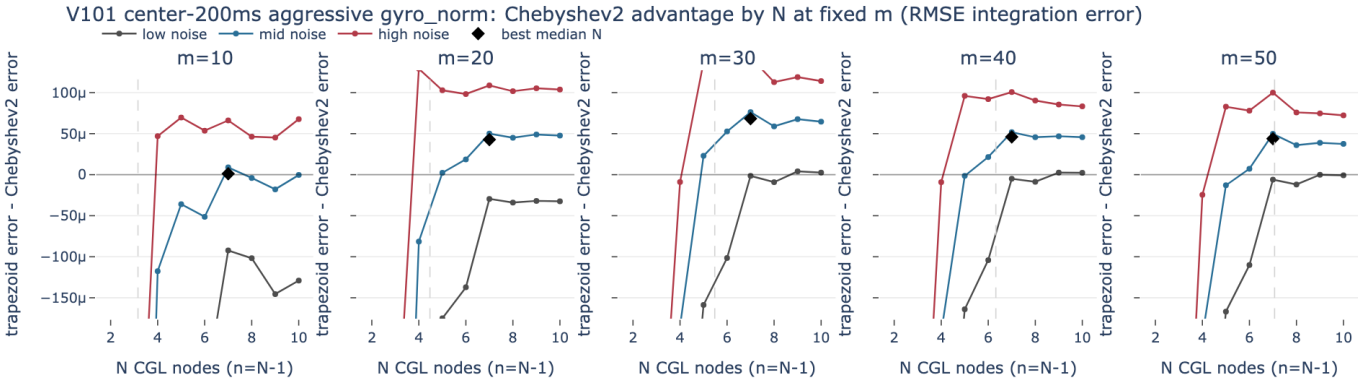


Figure 62: RMSE advantage curves for `V101 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V101 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	8	7	10	10	0%	50%	100%	-9.77e-06	.=%@@@@@@
20	4.47	2	7	9	9	25%	100%	100%	4.2e-05	.=%@@@@@@
30	5.48	5	7	7	7	50%	100%	100%	6.829e-05	.=%@@@@@@
40	6.32	2	7	7	7	50%	100%	100%	4.602e-05	.=%@@@@@@
50	7.07	2	7	7	9	50%	100%	100%	3.457e-05	.=%@@@@@@

Figure 63: Robust `N` table for `V101 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.7 `V102 center-200ms aggressive gyro_norm`

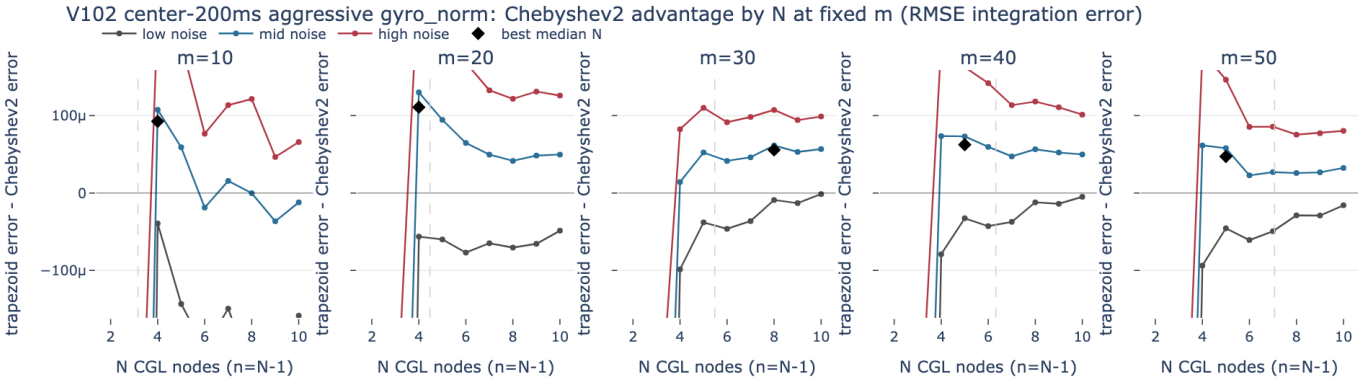


Figure 64: RMSE advantage curves for `V102 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V102 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	7	4	4	5	0%	75%	100%	4.195e-05	.-@@@@@@@
20	4.47	3	4	4	4	25%	100%	100%	0.0001109	.-@@@@@@@
30	5.48	7	8	8	8	50%	100%	100%	5.564e-05	.-@@@@@@@
40	6.32	4	5	4	4	0%	100%	100%	5.873e-05	.-@@@@@@@
50	7.07	3	5	5	9	0%	100%	100%	2.073e-05	.-@@@@@@@

Figure 65: Robust `N` table for `V102 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.8 `V103 center-200ms aggressive gyro_norm`

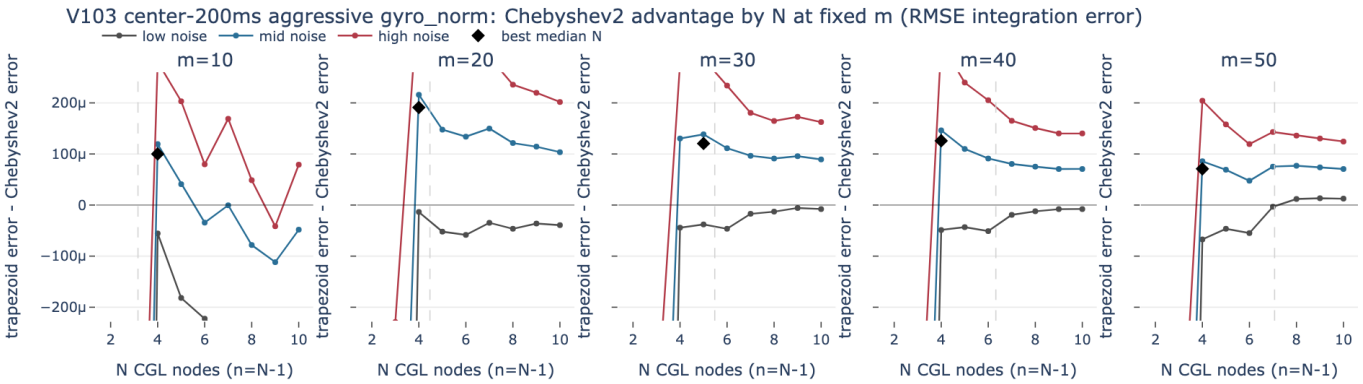


Figure 66: RMSE advantage curves for `V103 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V103 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	4	4	5	0%	75%	100%	2.01e-05	.=@@@@@@@
20	4.47	4	4	4	4	50%	100%	100%	0.0001911	.+@@@@@@@
30	5.48	5	5	4	5	50%	100%	100%	0.0001206	.+@@@@@@@
40	6.32	4	4	4	4	50%	100%	100%	0.0001259	.=@@@@@@@
50	7.07	10	4	4	4	0%	100%	100%	7.089e-05	.=@@@@@@@

Figure 67: Robust `N` table for `V103 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.9 `V201 center-200ms aggressive gyro_norm`

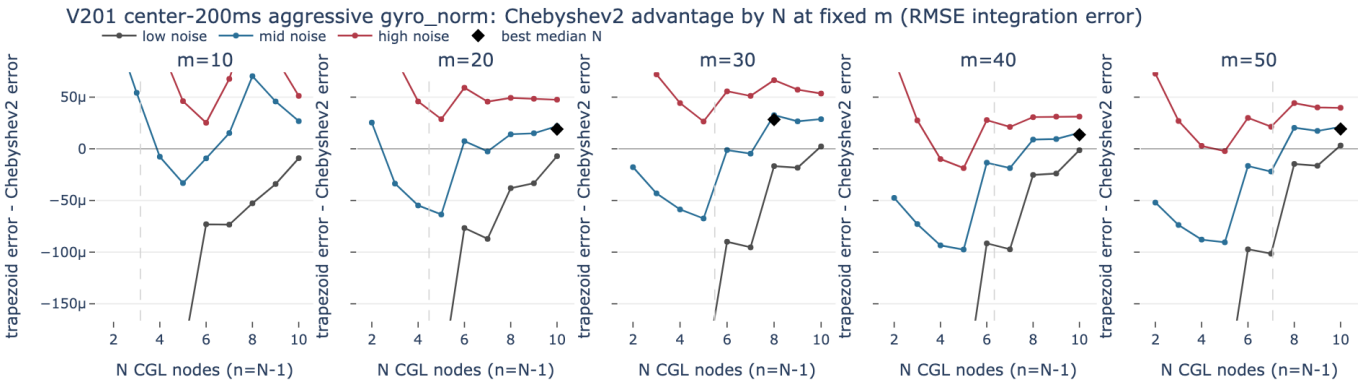


Figure 68: RMSE advantage curves for `V201 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V201 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	7	2	8	2	0%	75%	100%	8.906e-05	@+_.:=#×+
20	4.47	2	10	10	10	50%	100%	100%	1.899e-05	#:_.#×%%@
30	5.48	5	8	10	8	25%	100%	100%	2.827e-05	=:_.××@@@
40	6.32	10	10	10	10	50%	100%	100%	1.36e-05	-:..#×%%@
50	7.07	5	10	10	8	25%	100%	100%	1.733e-05	-_..××@@@

Figure 69: Robust `N` table for `V201 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.10 `V202 center-200ms aggressive gyro_norm`

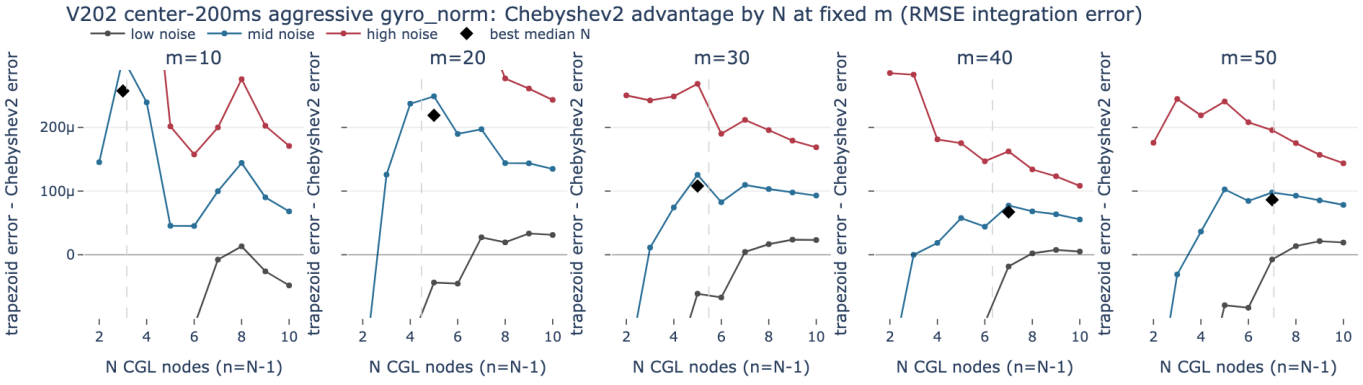


Figure 70: RMSE advantage curves for `V202 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V202 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	3	3	3	25%	100%	100%	0.0002572	.@#..:=:_
20	4.47	3	5	5	4	25%	100%	100%	0.0001981	.#@@%###
30	5.48	5	5	5	5	25%	100%	100%	0.0001079	.*%@%@@@%
40	6.32	7	7	8	8	50%	100%	100%	6.061e-05	.*#%%@@@%
50	7.07	7	7	9	7	50%	100%	100%	8.626e-05	.*#@@@@@

Figure 71: Robust `N` table for `V202 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.11 `V203 center-200ms aggressive gyro_norm`

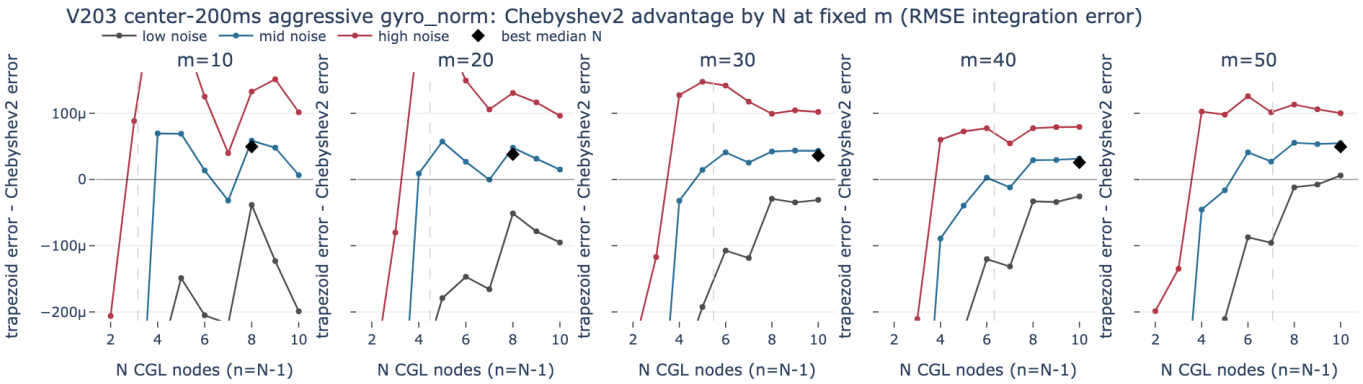


Figure 72: RMSE advantage curves for `V203 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V203 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	8	8	5	0%	75%	100%	4.856e-05	.+@@@%@@@
20	4.47	5	8	8	8	0%	100%	100%	3.796e-05	.=@@@@@@@
30	5.48	5	10	8	8	25%	100%	100%	3.54e-05	.-%@@@@@@
40	6.32	5	10	10	10	0%	100%	100%	2.579e-05	.-%%@@@@@
50	7.07	7	10	10	10	50%	100%	100%	4.943e-05	.-%%@@@@@

Figure 73: Robust `N` table for `V203 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).